

Noun–Verb Analysis

MOVIE TICKET BOOKING SYSTEM

1. Nouns Identified

The following nouns are identified from the problem statement and used as candidate classes or design concepts:

Movie, Ticket, Booking System, Cinema, Customer, Show, Seat, Payment, Booking, Screen, Seat Type, Ticket

2. Verbs Identified

The following verbs/actions are identified from the problem statement:

build, see, play, pick, book, pay, get, cancel, display, calculate, print

3. Noun–Verb Analysis Table

Noun Found	Keep as a Class?	Reason
Cinema	Yes	Represents the theatre and owns its screens.
Screen	Yes	Represents one auditorium and owns its physical seats.
Movie	Yes	Has its own identity and data such as title, language and duration.
Show	Yes	Represents one screening of a movie on a particular screen at a particular time.
Seat	Yes	Represents a physical seat with a seat number and seat type.
ShowSeat	Yes	Represents the status of one physical seat for one particular show.
Customer	Yes	Represents the customer and stores customer information.
Booking	Yes	Represents a customer's booking, including booking ID, seats, amount and status.

Payment	Yes	Represents the payment abstraction and defines the payment contract.
UPI Payment	Yes	Represents one concrete payment method.
Card Payment	Yes	Represents one concrete payment method.
Cash Payment	Yes	Represents one concrete payment method.
Ticket	No separate class	A ticket is the printable result of a successful booking; printing is handled by TicketPrinter.
Seat Layout	No	It is a view of a show's seats, not an independent entity; it can be displayed through a method.
Booking System	No separate class	The system is represented through the required service and menu classes, particularly BookingService and the main menu.
Seat Type	No separate class	It can be represented as a type/category such as SILVER, GOLD and PLATINUM within Seat.

4. Verb / Action to Class Responsibility Mapping

Verb / Action	Responsible Class
List movies	Cinema / Main Menu
List shows	Show / BookingService
Display seats	Show / ShowSeat
Check seat availability	ShowSeat
Book seats	BookingService / Booking
Calculate price	PriceCalculator
Pay	Payment
Pay using UPI	UpiPayment
Pay using Card	CardPayment
Pay using Cash	CashPayment
Print ticket	TicketPrinter
Cancel booking	BookingService / Booking
Release seats	ShowSeat
Generate booking ID	Booking

5. Final Class Set

After the noun–verb analysis, the following classes are used for the system:

Category	Classes
Core Entity Classes	Cinema, Screen, Seat, Movie, Show, ShowSeat, Customer, Booking
Payment Classes	Payment, UpiPayment, CardPayment, CashPayment
Service / Behaviour Classes	PriceCalculator, TicketPrinter, BookingService
Program / Interface	Main / Cinema Menu

6. Key Design Decisions

Not every noun is converted into a class. A noun should become a class when it has its own identity, data, behaviour, or responsibility in the system.

Seat and ShowSeat are kept separate. Seat represents the physical seat, while ShowSeat represents the status of that seat for one particular show.

Seat Layout is not a separate class because it is a view of ShowSeat objects.

Ticket is not treated as a separate entity class because the assignment assigns ticket formatting and printing to TicketPrinter.